

Exercise: let $x_0 = 1$, $x_n = T^n(x_0)$ [This goes to ∞]

Corollary (to CM1): let (X, d) be complete metric space, $T: X \rightarrow X$

satisfies that for some $n_0 \in \mathbb{N}$,

$$T^{n_0} = \underbrace{T \circ T \circ \dots \circ T}_{n_0\text{-fold composition}} : X \rightarrow X$$

is a strict contraction. Then

- $\exists!$ fixed point $x_{f,x}$ for T
- if $x_0 \in X$, $x_n = T^n(x_0) \xrightarrow{n \rightarrow \infty} x_{f,x}$

proof: By CM1, T^{n_0} admits a unique fixed point.

$$\text{Then } T^{n_0}(T(x_{f,x})) = T^{n_0+1}(x_{f,x}) = T(T^{n_0}(x_{f,x})) = T(x_{f,x})$$

So $T(x_{f,x})$ is the fixed point of T^{n_0} .

By uniqueness, $T(x_{f,x}) = x_{f,x}$

let $r = \{0, 1, \dots, n_0-1\}$, and $n_k^{(r)} = kn_0 + r$, $k \in \mathbb{N}$

Then, for $x_0 \in X$, $T^{n_k^{(r)}}(x_0) = (T^{n_0})^k (T^r(x_0)) \xrightarrow{k \rightarrow \infty} x_{f,r}$
 $\uparrow$ T^0 : id map

by applying part (ii) of CT to T^{n_0} .

However, the subsequence $(T^{n_k^{(0)}}(x_0))_{k=1}^{\infty}, \dots, (T^{n_k^{(n_0-1)}}(x_0))_{k=1}^{\infty}$

exists the sequence $(T^n(x_0))_{n=1}^{\infty}$.

Thus, $\lim_{n \rightarrow \infty} T^n(x_0) = x_{f,x}$

□.

Global Picard's Theorem

let $a < b$ in $\mathbb{R}$. Consider the initial value problem

(IVP) : $f'(t) = \Phi(t, f(t))$, $f(t_0) = y_0$; $t_0 \in [a, b]$, $y_0 \in \mathbb{R}$

where $\Phi: [a, b] \times \mathbb{R} \rightarrow \mathbb{R}$ satisfies that it is

• cts on first variable: $t \mapsto \Phi(t, y)$ cts, $\forall y \in \mathbb{R}$.

• Lipschitz in second variable: $\forall t \in [a, b]$,

$$|\Phi(t, y_1) - \Phi(t, y_2)| \leq \underset{\substack{\uparrow \\ \text{universal}}}{L} |y_1 - y_2| \quad \text{for } y_1, y_2 \in \mathbb{R}.$$

Then (IvP) has unique sol'n in $C[a, b]$

(further, f is differentiable)

proof (I): let's set up an integral equation.

We apply F.T. of C. to get

$$\underbrace{f(t) - f(t_0)}_{\gamma_0} = \int_{t_0}^t f'(s) ds = \int_{t_0}^t \Phi(s, f(s)) ds$$

$$\Leftrightarrow f(t) = \gamma_0 + \int_{t_0}^t \Phi(s, f(s)) ds \stackrel{\text{def'n}}{=} T(f)(t) \quad (IE)$$

(II). let $T: C[a, b] \rightarrow C[a, b]$ defined to be

$$T(f)(t) = \gamma_0 + \int_{t_0}^t \Phi(s, f(s)) ds$$

Note that the "antiderivative" fn is always cts.

[Fact: $s \mapsto \bar{\Phi}(s, f(s))$ is cts, so $T(f)$ is differentiable]

Consider for $f, g \in C[a, b]$, and $t \in [a, b]$

$$|Tf(t) - Tg(t)| = \left| \gamma_0 + \int_{t_0}^t \bar{\Phi}(s, f(s)) ds - \gamma_0 - \int_{t_0}^t \bar{\Phi}(s, g(s)) ds \right|$$

$$= \left| \int_{t_0}^t [\bar{\Phi}(s, f(s)) - \bar{\Phi}(s, g(s))] ds \right|$$

wlog,

say $t \geq t_0$

$$\leadsto \leq \int_{t_0}^t |\bar{\Phi}(s, f(s)) - \bar{\Phi}(s, g(s))| ds$$

$$\leq \int_{t_0}^t L \underbrace{|f(s) - g(s)|}_{\leq \|f-g\|_{\infty}} ds$$

$$\leq \int_{t_0}^t L \underbrace{\|f-g\|_{\infty}}_{\text{constant}} ds$$

$$= (t - t_0) L \|f-g\|_{\infty}.$$

we shall prove for $n \in \mathbb{N}$,

$$|T^n f(t) - T^n g(t)| \leq \frac{L^n \|f-g\|_{\infty}}{n!} \cdot |t-t_0|^n \quad (*)$$

Case 1: if $n=1$, then done.

Assuming $(*)$ holds for n , consider

$$\begin{aligned} & |\bar{T}^{n+1} f(t) - \bar{T}^{n+1} g(t)| \\ &= \left| \int_{t_0}^t [\bar{\Phi}(s, \bar{T}^n f(s)) - \bar{\Phi}(s, \bar{T}^n g(s))] ds \right| \end{aligned}$$

$$\stackrel{\text{wlog,}}{t > t_0} \leq \int_{t_0}^t |\bar{\Phi}(s, \bar{T}^n f(s)) - \bar{\Phi}(s, \bar{T}^n g(s))| ds$$

$$\leq \int_{t_0}^t L \cdot |\bar{T}^n f(s) - \bar{T}^n g(s)| ds$$

$$\leq \int_{t_0}^t L \cdot \frac{L^n \|f-g\|_{\infty}}{n!} \cdot |s-t_0|^n ds, \text{ inductive hypothesis.}$$

$$= \frac{L^{n+1} \|f-g\|_{\infty}}{(n+1)!} (t-t_0)^{n+1}$$

(IV). $\bar{T} \text{ fixed } \varepsilon$.

$$\text{Now, let } n_0 \in \mathbb{N} \text{ s.t. } \frac{L^{n_0} (b-a)^{n_0}}{n_0!} < 1$$

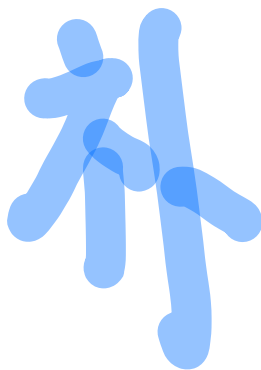
Then for $f, g \in C[a, b]$, we have

$$\begin{aligned} |T^{n_0} f(t) - T^{n_0} g(t)| &\leq \frac{L^{n_0} \cdot \|f-g\|_{\infty}}{n_0!} \cdot \underbrace{|t-t_0|^{n_0}}_{\leq (b-a)^{n_0}} \\ &\leq \frac{L^{n_0} \cdot (b-a)^{n_0}}{n_0!} \cdot \underbrace{\|f-g\|_{\infty}}_{L > d_c(f, g)} \end{aligned}$$

$$\Rightarrow \|T^{n_0} f(t) - T^{n_0} g(t)\|_{\infty} \leq c \cdot \|f-g\|_{\infty}$$

So we use (Corollary to) CIVT

□.



Corollary (Banach - Steinhaus Theorem)

let $(V, \|\cdot\|_V)$, $(W, \|\cdot\|_W)$ be normed spaces.

with $(V, \|\cdot\|_V)$ complete. let

$$\{T_i\}_{i \in I} \subset B(V, W) \quad \left(\begin{array}{l} \text{cts linear operator from} \\ V \text{ to } W, \text{ "cts" = "bounded"} \end{array} \right)$$

be pointwise bounded:

$$\text{for each such } v, \sup_{i \in I} \|T_i v\|_W < \infty.$$

then $\sup_{i \in I} \|T_i\| < \infty$ (uniform boundedness of Lipschitz constant)

Recall: $T: V \rightarrow W$ is cts. then

$$\|T\| = \sup_{v \in B_V[0,1]} \|Tv\|_W \quad \left\{ \begin{array}{l} \cdot \text{ Lipschitz constant} \\ \cdot \text{ uniform bound on } B_V[0,1] \end{array} \right.$$

proof: for each i in I , let $f_i(v) = \|T_i v\|_W$. so

$f_i \in C(V)$ (composition of cts T with cts norm f_W)

Then the assumption that

$$\forall v \in V, \sup_{i \in I} |f_i(v)| = \sup_{i \in I} \|T_i v\|_W < \infty$$

tells us that there is non-empty open $U \subset V$.

So

$$\sup_{i \in I} \sup_{u \in U} |f_i(u)| \leq M < \infty \quad (\text{Some } M > 0)$$

let $x \in U, \varepsilon > 0$ be so $B_V[x, \varepsilon] \subset U$ (!! comment before BC)

If $z \in B_V[0, 1]$, we have

$$z = \frac{1}{2\varepsilon} (-x_0 + \varepsilon z) + \frac{1}{2\varepsilon} (x_0 + \varepsilon z)$$

where $x_0 \pm \varepsilon z \in B_V[x_0, \varepsilon] \subset U$. So for each $i \in I$,

$$\begin{aligned} \|T_i z\|_W &\leq \frac{1}{2\varepsilon} \|T_i(x_0 - \varepsilon z)\|_W + \frac{1}{2\varepsilon} \|T_i(x_0 + \varepsilon z)\|_W \\ &\leq \frac{M}{\varepsilon} \end{aligned}$$

Hence $\|T_i\| = \sup_{z \in B_V[0, 1]} \|T_i z\|_W \leq \frac{M}{\varepsilon}$

So $\sup_{i \in I} \|T_i\| \leq \frac{M}{\varepsilon} < \infty$

□.

Remark. If $f \in C([0, 2\pi])$, $S_n f(t) = a_0 + \sum_{k=1}^n [a_k \cos(kt) + b_k \sin(kt)]$

$$a_k = \frac{1}{\sqrt{2}\pi} \int_0^{2\pi} f(s) \cos(ks) ds \quad ; \quad b_k = \frac{1}{\sqrt{2}\pi} \int_0^{2\pi} f(s) \sin(ks) ds$$

Ex: $S_n: C[0, 2\pi] \rightarrow C[0, 2\pi]$ is a linear operator

with $\|S_n\| \geq C \log n$ [p. 450]

$$\text{so } \sup_{n \in \mathbb{N}} \|S_n\| = \infty.$$

By contrapositive of B-S Thm, there is $f \in C[0, 2\pi]$ so

$$\sup_{n \in \mathbb{N}} \|S_n f\|_\infty = \infty \Rightarrow \lim_{n \rightarrow \infty} S_n f \text{ D.M.B. in } C[0, 2\pi].$$

On spaces $C(X)$, (X, d) compact. $C(X) = \{f \in \mathbb{R}^X : f \text{ is cts}\}$

EUT $\Rightarrow$ there is $x_{\max}$ in X , so $|f(x)| \leq |f(x_{\max})|$

$$\text{so } \|f\|_\infty = \sup_{x \in X} |f(x)| < \infty = \max_{x \in X} |f(x)| = |f(x_{\max})|$$

Our major normed space $(C(X), \|\cdot\|_\infty)$

Recall: this is complete.

$$\begin{cases} d_c \text{ on } C_b(X, Y) \\ \text{if } X \text{ is compact, } C_b(X, \mathbb{R}) = C(X, \mathbb{R}) = C(X). \end{cases}$$

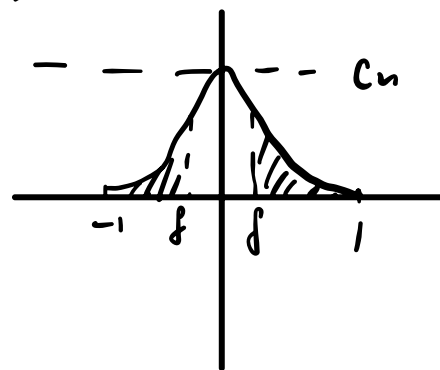
Weierstrass Approximation Theorem.

Let $a < b$ in $\mathbb{R}$. Then the set of polynomial functions $P[a, b]$ is uniformly dense in $C[a, b]$, i.e.,

$$\overline{P[a, b]} = C[a, b] \text{ in } (C[a, b], \|\cdot\|_\infty)$$

Lemma: For n in $\mathbb{N}$, let $f_n(t) = C_n(1-t^2)^n$,

where $C_n = \frac{1}{\int_0^1 (1-t^2)^n dt}$



Then (f1): $f_n(t) \geq 0, \forall t \in [-1, 1]$

(f2): $\int_0^1 f_n(s) ds = 1, \forall n \in \mathbb{N}$

(f3): $0 \leq \left(\int_{-1}^{-\delta} + \int_{\delta}^1 \right) f_n(s) ds \xrightarrow{n \rightarrow \infty} 0, \forall \delta \in (0, 1]$

Proof: (f1) obvious, (f2) by choice of C_n

(f3) We use that $0 \leq t^2 \leq t \leq 1, \forall t \in [0, 1]$, to get

$$\begin{aligned} \frac{1}{c_n} &= \int_0^1 (1-t^2)^n dt \geq \int_0^1 (1-t)^n dt \\ &= \frac{(1-t)^{n+1}}{n+1} \Big|_0^1 = \frac{1}{n+1} \end{aligned}$$

and hence $c_n \leq n+1$. Then by symmetry,

$$\begin{aligned} 0 &\stackrel{\text{by (8.1)}}{\leq} \left(\int_{-1}^{-\delta} + \int_{\delta}^1 \right) \tilde{p}_n(t) dt = 2 \int_{\delta}^1 \tilde{p}_n(t) dt \\ &= 2 \int_{\delta}^1 c_n (1-t^2)^n dt \\ &\leq 2 \int_{\delta}^1 (n+1) (1-\delta^2)^n dt \\ &= 2(n+1) (1-\delta^2)^n (1-\delta) \xrightarrow{n \rightarrow \infty} 0 \quad \square. \end{aligned}$$

proof: (W. Approx. Thm):

(I). let $[a, b] = [0, 1]$, and $f \in C[0, 1]$ with $f(0) = 0 = f(1)$

By letting $f(t) = 0$ if $t \in (-\infty, 0) \cup (1, +\infty)$, we may

assume $f \in C_b(\mathbb{R})$

let's define for $n \in \mathbb{N}$, $p_n(t) = \int_0^1 \tilde{p}_n(s-t) f(s) ds$, $t \in [0, 1]$

Then p_n is a polynomial, $f_n(s-t) = \sum_{k=0}^{2n} \sum_{j=0}^{2n} a_{k,j} \cdot t^k \cdot s^j$

$$\text{Then } p_n(t) = \sum_{k=0}^{2n} \left[\sum_{j=0}^k a_{k,j} \cdot \int_0^1 s^j f(s) ds \right] \cdot t^k$$

which is a polynomial function.

$$\text{Then we let } u = s-t, \text{ so } p_n(t) = \int_{-t}^{1-t} f_n(u) f(u+t) du$$

We consider for $t \in [0,1]$,

$$\begin{aligned} |p_n(t) - f(t)| &= \left| \int_{-t}^{1-t} f_n(u) f(u+t) du - \int_0^1 \overbrace{f_n(u) f(t)}^{(f_2)} du \right| \\ &= \left| \int_{-1}^1 f_n(u) f(u+t) du - \int_{-1}^1 f_n(u) f(t) du \right| \quad (\text{by choice of } f) \end{aligned}$$

$$\leq \int_{-1}^1 \underbrace{f_n(u)}_{(f_1)} |f(u+t) - f(t)| dt$$

let $\varepsilon > 0$, let $\delta > 0$ be so $|s-t| < \delta \Rightarrow |f(s) - f(t)| < \frac{\varepsilon}{2}$ (unif. cont. of f)

• $[0,1]$ compact $\Rightarrow f$ is unif. cont. on $[0,1]$.

• $f=0$ off of $[0,1]$

Thus, for $\delta' \in (0, \delta)$,

$$\begin{aligned}
 |p_n(t) - f(t)| &\leq \int_{-1}^1 \delta_n(u) |f(u+t) - f(t)| du \\
 &\leq \left(\int_{-1}^{-\delta'} + \int_{\delta'}^1 \right) \delta_n(u) \underbrace{|f(u+t) - f(t)|}_{\leq \|f\|_\infty} du \\
 &\quad + \underbrace{\int_{-\delta'}^{\delta'} \delta_n(u) |f(u+t) - f(t)| du}_{< \frac{\varepsilon}{2}, \text{ by unif. cty.}}
 \end{aligned}$$

$$< 2\|f\|_\infty \left(\int_{-1}^{-\delta'} + \int_{\delta'}^1 \right) \delta_n(u) du$$

$$+ \frac{\varepsilon}{2} \cdot \underbrace{\int_{-1}^1 \delta_n(u) du}_{\text{using (f1) to bound up an integral.}}$$

$$\xrightarrow{n \rightarrow \infty} \underbrace{0}_{(f3)} + \frac{\varepsilon}{2} \cdot \underbrace{2}_{(f2)} = \varepsilon$$

and hence there is a $n_\varepsilon \in \mathbb{N}$ so $n \geq n_\varepsilon$

$$\Rightarrow |p_n(t) - f(t)| < \varepsilon$$

and estimates are independent of choice of t .

$$\text{i.e. } \|p_n - f\|_\infty < \varepsilon, \forall n > n_\varepsilon$$

$$\text{so } \lim_{n \rightarrow \infty} \|p_n - f\|_\infty = 0 \quad (\text{unif. conv})$$

(II). Now consider $a < b$, $f \in C[a, b]$

Define for $t \in [0, 1]$.

$$Tf(t) = f(a + (b-a)t) - [f(b) - f(a)]t - f(a)$$

$$\text{so if } f \in C[0, 1], Tf(0) = 0, \text{ and } Tf(1) = 0$$

Then (I) provides a sequence $(p_n)_{n=1}^\infty$ of polynomial functions so

$$\lim_{n \rightarrow \infty} \|p_n - f\|_\infty = 0 \quad \text{in } (C[0, 1], \|\cdot\|_\infty)$$

if $\tilde{p}_n(s)$ is a polynomial so

$$\tilde{p}_n(a + (b-a)t) - [f(b) - f(a)]t - f(a) = p_n(t)$$

Then we have $\max_{s \in [a,b]} |\tilde{p}_n(s) - f(s)|$

$$= \max_{t \in [0,1]} |p_n(t) - Tf(t)| \xrightarrow{n \rightarrow \infty} 0$$

$$\text{So } \|\tilde{p}_n - f\|_{\infty} \xrightarrow{n \rightarrow \infty} 0 \text{ in } (C[a,b], \|\cdot\|_{\infty})$$

□.

Theorem (Nowhere differentiable functions are generic)

Let $ND[0,1] = \{f \in C[0,1] : f \text{ is differentiable at no point}\}$

Then $ND[0,1]$ is residual in $(C[0,1], \|\cdot\|_{\infty})$

proof: For $M, \delta > 0$, $F_{\delta, M} = \{f \in C[0,1] : L_{\delta, M}(f) \neq \emptyset\}$

where $L_{\delta, M}(f) = \{x \in [0,1] : |f(t) - f(x)| \leq M|t - x|, \text{ provided } t \in (x - \delta, x + \delta)\}$

"Lipschitz near some x "

Note (i): $M' \geq M$, $\delta' \leq \delta$ $M(0, +\infty)$, then

$$F_{M, \delta} \subseteq F_{M', \delta'}$$

vi). If f is differentiable at x , then

$$\lim_{\substack{t \rightarrow x \\ t \neq x}} \frac{|f(t) - f(x)|}{|t - x|} = |f'(x)|$$

and if $M = |f'(x)| + 1$, then for some $\delta > 0$,

$$x \in L_{M, \delta}(f), \text{ i.e., } f \in F_{M, \delta}$$

(I). Let us see that $F_{M, \delta}$ is closed

Let $(f_n)_{n=1}^{\infty} \in (F_{M, \delta})^{\mathbb{N}}$ have uniform limit.

$f = \lim_{n \rightarrow \infty} f_n$. Each $L_M(f_n) \neq \emptyset$ so we have

$x_n \in L_{M, \delta}(f_n)$. Since $[0, 1]$ is compact, we can

extract a convergent subseq $x = \lim_{k \rightarrow \infty} x_{n_k} \in [0, 1]$

Now, if $t \in B(x, \delta)$, and let $k_{\delta} \in \mathbb{N}$ be so

$$|x - x_{n_k}| < \delta - |t - x|, \text{ for } k \geq k_{\delta}.$$

hence

$$|t - x_{n_k}| < |t - x| + |x - x_{n_k}| < \delta, \text{ for } k \geq k_{\delta}.$$

$$\begin{aligned}
\text{So } |f(t) - f(x)| &\leq |f(t) - f_{n_k}(t)| + |f_{n_k}(t) - f_{n_k}(x_{n_k})| \\
&\quad + |f_{n_k}(x_{n_k}) - f(x_{n_k})| + |f(x_{n_k}) - f(x)| \\
&\leq |f(t) - f_{n_k}(t)| + \underbrace{M|t - x_{n_k}|}_{t \in B(x_{n_k}, \delta)} + |f_{n_k}(x_{n_k}) - f(x_{n_k})| \\
&\quad + \underbrace{M|x_{n_k} - x|}_{x \in B(x_{n_k}, \delta)} \\
&\xrightarrow{n \rightarrow \infty} 0 + M|t - x| + 0 + 0, \text{ by uniform c.p.} \\
&\quad \text{and } x_{n_k} \xrightarrow{n \rightarrow \infty} x \\
&= M|t - x|
\end{aligned}$$

i.e., $|f(t) - f(x)| \leq M|t - x|$, so $x \in L_{M, \delta}(f)$, so $f \in F_{M, \delta}$

(II). Let us see that $F_{M, \delta} = \emptyset$ in $(C[a, b], \|\cdot\|_\infty)$

i.e., nowhere dense.

Let $f \in F_{M, \delta}$, $\varepsilon > 0$. we will see that

$B(f, \varepsilon) \not\subset F_{M, \delta}$ true for any $\varepsilon > 0$.

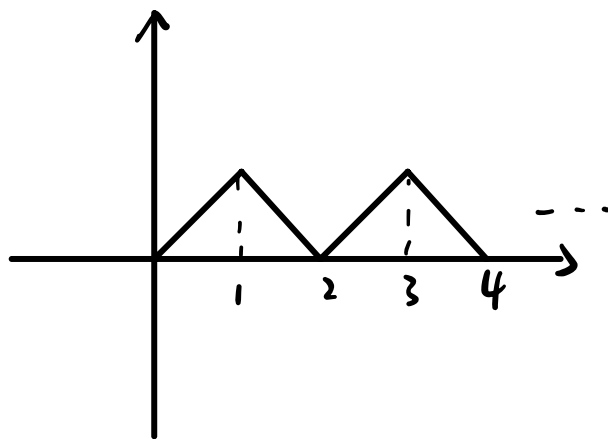
[idea: approach with good, ..., the good with bad].

First, by W.A.I., we have a polynomial function p on $[0,1]$.

$$\text{so } \|f - p\|_{\infty} < \frac{\varepsilon}{2}.$$

$$\text{let } M' = \max_{t \in [0,1]} |p'(t)|$$

$$\text{let } \varphi: [0, +\infty) \rightarrow [0,1], \varphi(t) = \begin{cases} t-n, & t \in [n, n+1], n \in \mathbb{N} \cup \{0\}, \text{ even} \\ n-t, & t \in [n, n+1], n \in \mathbb{N}, \text{ odd.} \end{cases}$$



$$\text{For } k \in \mathbb{N}, \text{ let } \varphi_k(t) = \frac{1}{k} \varphi(k^2 t)$$

$$\text{so if } s, t \in \left[\frac{n}{k^2}, \frac{n+1}{k^2}\right], n \in \{0\} \cup \mathbb{N}, \text{ then}$$

$$|\varphi_k(s) - \varphi_k(t)| = k |s - t|$$

Now we consider $k \in \mathbb{N}$,

$$\frac{1}{k} < \frac{\varepsilon}{2}, \quad \frac{1}{k^2} < \delta, \quad \text{and } k - M' > M.$$

and let $x \in (0,1)$, then for s in $(0,1)$ so

$s, x \in [\frac{n}{k^2}, \frac{n+1}{k^2}]$, we have for $\psi_k = \varphi_k + p$.

we have

$$|\psi_k(s) - \psi_k(x)| = |\varphi_k(s) + p(s) - \varphi_k(x) - p(x)|$$

$$\geq |\varphi_k(s) - \varphi_k(x)| - |p(s) - p(x)|$$

$$\geq k|s-x| - \underbrace{|p'(s-x)|}_{\leq M'}|s-x|$$

$$\geq (k - M')|s-x| \geq M|s-x|$$

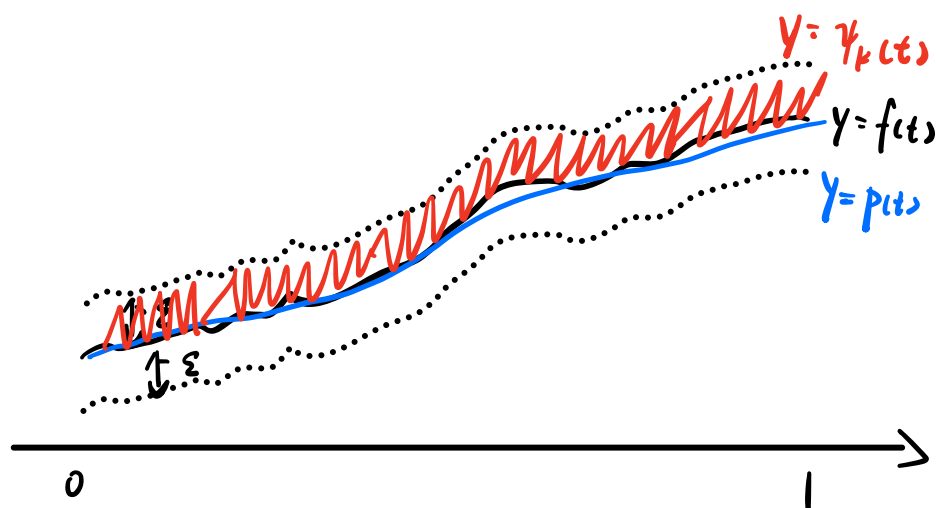
$$\text{and } \|\psi_k - f\|_\infty \leq \|\psi_k - p\|_\infty + \|p - f\|_\infty$$

$$= \|\varphi_k\|_\infty + \|p - f\|_\infty$$

$$< \frac{1}{k} + \underbrace{\frac{\varepsilon}{2}}_{\text{choice of } p} < \varepsilon$$

Hence $L_{M,\delta}(\psi_k) = \emptyset$, so $\psi_k \notin F_{M,\delta}$ but

$$\psi_k \in B(f, \varepsilon)$$



(IV). Finally, we let

$$SD[0,1] = C[0,1] \setminus ND[0,1]$$

We saw in (ii), above, that if $f \in SD[0,1]$,

$f \in F_{M,\delta}$ for some $M, \delta > 0$, i.e.,

$$SD[0,1] \subseteq \bigcup_{(M,\delta) \in (0,\infty)^2} F_{M,\delta}$$

By (i), above, if $n \in \mathbb{N}$ is so large that

$$n \geq M, \frac{1}{n} < \delta, \text{ then } F_{M,\delta} \subseteq F_{n,\frac{1}{n}}.$$

$$\text{Thus, } SD[0,1] \subseteq \bigcup_{n=1}^{\infty} F_{n,\frac{1}{n}}$$

where each $F_{n,\frac{1}{n}}$ is N.D., so $SD[0,1]$ is meager $\square$.

Recall: (X, d) compact metric space, $(C(X), \|\cdot\|_\infty)$
complete metric space.

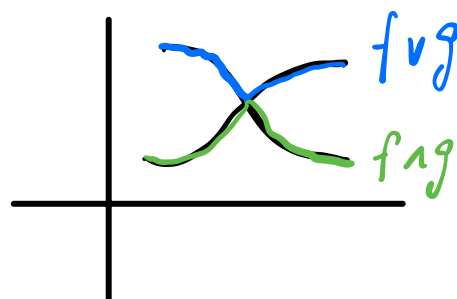
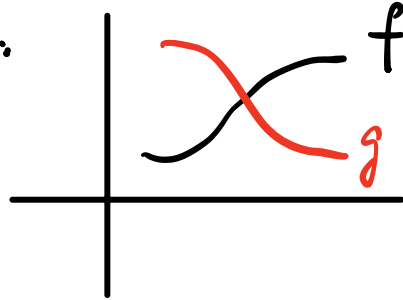
Lattice Structure in $C(X)$:

If $f, g \in C(X)$, let

$$f \vee g(x) = \max\{f(x), g(x)\}, x \in X$$

$$f \wedge g(x) = \min\{f(x), g(x)\}, x \in X.$$

Picture:



Remark: Each of $f \vee g, f \wedge g \in C(X)$ if $f, g \in C(X)$

If $a, b \in \mathbb{R}$, $\max\{a, b\} = \frac{1}{2}(a+b) + \frac{1}{2}|a-b|$, hence

$$f \wedge g = \frac{1}{2} (f + g - |f - g|)$$

$$f \vee g = -[(-f) \wedge (-g)]$$

Associativity: $f, g, h \in C(X)$, $(f \vee g) \vee h = f \vee (g \vee h)$

So we write $f \wedge g \wedge h$, similar with $f \vee g \vee h$

Stone's Approximation Theorem:

Let (X, d) be a compact metric space.

Let $L \subseteq C(X)$ be s.t.

- L is a $\mathbb{R}$ -subspace.
- L is a sublattice. $f, g \in L \Rightarrow f \wedge g, f \vee g \in L$.
- L is point separating: if $x \neq y \in X$, there is $f \in L$ s.t. $f(x) \neq f(y)$.

$$\cdot 1 \in \mathcal{L}, \quad 1(x) = 1 \quad \text{for } x \in X.$$

$$\text{Then } \overline{\mathcal{L}} = C(X), \text{ in } (C(X), \|\cdot\|_\infty)$$

proof: (i): If $x \neq y \in \mathbb{R}$, $\alpha, \beta \in \mathbb{R}$.

then there is $g \in \mathcal{L}$ so $g(x) = \alpha, g(y) = \beta$.

To prove this, use point-separating to find $\varphi \in \mathcal{L}$

s.t. $\varphi(x) \neq \varphi(y)$. Let

$$g = \alpha \cdot 1 + \frac{\beta - \alpha}{\varphi(y) - \varphi(x)} [\varphi - \varphi(x) \cdot 1]$$

Now, before we proceed, fix $f \in C(X)$, $\varepsilon > 0$.

(1). Fix $x \in X$. We will find $g_x \in \mathcal{L}$ s.t.

$$\underbrace{g_x(x)}_{\text{Value at one point}} = f(x) \quad \text{and} \quad g_x < f + \varepsilon \cdot 1.$$

To do this, let $y \in X$, and use (a) with

$$\alpha = f(x), \quad \beta = f(y) \quad \text{to find} \quad [\alpha = \beta = f(x) \Rightarrow x = y]$$

$g_{x,y}$ in $\mathcal{L}$ with $g_{x,y}(x) = f(x)$, $g_{x,y}(y) = f(y)$.

Then, for each y in X , there is $\delta_y > 0$

$$[g_{x,y}(y) - f(y) = 0, \text{ each } g_{x,y}, f \text{ cts}]$$

$$\text{so} \quad d(y, z) < \delta_y \Rightarrow g_{x,y}(z) - f(z) < \varepsilon.$$

$$\Rightarrow g_{x,y}(z) < f(z) + \varepsilon$$

Then $X = \bigcup_{y \in X} B(y, \delta_y)$, and compactness provides that

$$X = \bigcup_{j=1}^n B(y_j, \delta_{y_j}) \quad \text{for some } y_1, \dots, y_n \in X.$$

$$\text{let} \quad g_x = g_{x,y_1} \wedge \dots \wedge g_{x,y_n} \in \mathcal{L}.$$

Now, if $z \in X$, $z \in B(y_j, \delta y_j)$ for some $j \in \{1, \dots, n\}$

$$\text{so } g_X(z) \leq g_{X, y_j}(z) < f(z) + \varepsilon$$

$$\Rightarrow g_X < f + \varepsilon \cdot 1, \text{ and } g_X(x) = g_{X, y_j}(x) = f(x)$$

(II). Finalé. For each x in X , let g_x be as in (I).

Find for each x in X , $\delta_x > 0$ be so $[g_x(x) - f(x) = 0]$

$$d(z, x) < \delta_x \Rightarrow g_x(x) - f(x) > -\varepsilon$$

$$\Rightarrow g_X(z) > f(z) - \varepsilon.$$

Then $X = \bigcup_{x \in X} B(x, \delta_x)$ so, as before,

$$X = \bigcup_{k=1}^m B(x_k, \delta_{x_k}). \text{ Then we let}$$

$$g = g_{x_1} \vee \dots \vee g_{x_m}$$

As in proof of (I), we see

$$g > f - \varepsilon \quad 1.$$

Now, since each $g_{x_k} < f + \varepsilon \quad 1$, we also have

$$g < f + \varepsilon \quad 1.$$

Thus, in summary,

$$f - \varepsilon \quad 1 < g < f + \varepsilon \quad 1.$$

$$\Rightarrow \|f - g\|_{\infty} = \max_{x \in X} |f(x) - g(x)| < \varepsilon$$

$\Rightarrow g \in \mathcal{L} \cap B(f, \varepsilon)$. Hence, taking arbitrary $f \in C(X)$,

we see [by ball-meet criterion] that

$$\overline{\mathcal{L}} = C(X)$$

□.

Example (i): let $\mathcal{L} = \{f \in C[a,b] : f \text{ is piecewise affine}\}$

verify the four assumptions of Stone's Thm.

$$\Rightarrow \overline{\mathcal{L}} = C[a,b] \text{ in } (C[a,b], \|\cdot\|_\infty)$$

(ii): let C be the Cantor set,

$$\text{Then } \mathcal{L} = \{f \in C(C) : f(C) = \{a_1, \dots, a_n\}, n \in \mathbb{N}, a_1, \dots, a_n \in \mathbb{R}\}$$

verify the 4 assumptions of Stone's Thm,

$$\text{hence } \overline{\mathcal{L}} = C(C) \text{ in } (C(C), \|\cdot\|_\infty).$$

Def'n: let (X, d) be a compact metric space. An algebra

in $C(X)$ is a subset $\mathcal{A} \subseteq C(X)$ s.t.

if $f, g \in \mathcal{A}$, $\alpha \in \mathbb{R}$, then

$$\cdot f + \alpha g \in \mathcal{A}$$

$$\cdot f \cdot g \in \mathcal{A}$$

Generalizes: $P[a,b] \subset C[a,b]$

Note: if p is a polynomial, $f \in A$, $p(f) = a_0 1 + a_1 f + a_2 f^2 + \dots + a_n f^n \in A$.

where $p(t) = a_0 + a_1 t + \dots + a_n t^n$

Stone - Weierstrauss Theorem:

Suppose $A \subset C(X)$ satisfies

- A is an algebra
- A is point separating
- $1 \in A$.

Then $\bar{A} = C(X)$ in $(C(X), \|\cdot\|_\infty)$

Proof: (I). We show for $f \in A$, that $|f| \in \bar{A}$

let $a < b$ in $\mathbb{R}$ so $f(X) \subseteq [a,b]$.

(Using that $f(X)$ is compact, hence bounded in $\mathbb{R}$)

Then, W.A.T. provides a sequence $(p_n)_{n=1}^{\infty} \in P[a,b]^{\mathbb{N}}$

$$\text{s.t. } \|p_n - 1\|_{\infty} = \max_{t \in [a,b]} |p_n(t) - 1| \xrightarrow{n \rightarrow \infty} 0$$

$$\text{Then } \|p_n(f) - f\|_{\infty} = \max_{x \in X} |p_n(f(x)) - f(x)|$$

$$\leq \max_{t \in [a,b]} |p_n(t) - t|$$

$$= \|p_n - 1\|_{\infty} \xrightarrow{n \rightarrow \infty} 0$$

$$\text{So } f = \lim_{n \rightarrow \infty} \underbrace{p_n(f)}_{\in A} \in \bar{A} \quad \text{in } (C(X), \|\cdot\|_{\infty})$$

(II). Finalé. Since A is a linear subspace of $C(X)$,

so too is $\bar{A}$. Also, if $f, g \in \bar{A}$, then $f+g \in \bar{A}$ too.

Indeed, if $f = \lim_{n \rightarrow \infty} f_n$, $g = \lim_{n \rightarrow \infty} g_n$ in $(C(X), \|\cdot\|_{\infty})$

where each $f_n, g_n \in A$, then

$$\begin{aligned}
\|fg - f_n g_n\|_\infty &\leq \|fg - f g_n\|_\infty + \|f g_n - f_n g_n\|_\infty \\
&\leq \|f\|_\infty \cdot \|g - g_n\|_\infty + \|g_n\|_\infty \cdot \|f - f_n\|_\infty \\
&\quad \uparrow \quad \quad \quad \underbrace{\lim_{n \rightarrow \infty} g_n = g \Rightarrow (g_n)_{n=1}^\infty \text{ is bounded.}} \\
(\text{[} \|f \cdot g\|_\infty &\leq \|f\|_\infty \cdot \|g\|_\infty \text{]})
\end{aligned}$$

$$\leq \|f\|_\infty \cdot \|g - g_n\|_\infty + M \cdot \|f - f_n\|_\infty$$

$\uparrow \sup_{n \in \mathbb{N}} \|g_n\|_\infty$

$$\begin{array}{c}
n \rightarrow \infty \\
\hline
\longrightarrow 0
\end{array}$$

Thus, $\bar{A}$ is an algebra in $C(X)$.

By (I), if $f \in \bar{A}$, then $|f| \in \bar{A} = \overline{\bar{A}}$

Thus, if $f, g \in \bar{A}$, we have

$$f \vee g = \frac{1}{2}(f+g) - \frac{1}{2}(f-g) \in \bar{A}$$

and $f \wedge g = -[\frac{1}{2}(-f) \vee \frac{1}{2}(-g)] \in \bar{A}$ too.

Hence $\bar{A}$ is a lattice

• $\bar{A}$ is a linear subspace

• $1 \in A \subseteq \bar{A}$

• $\bar{A}$ is point separating, as $A \subseteq \bar{A}$

Thus by S.A.I., $\bar{A} = \overline{\bar{A}} = C(X)$

□.

Ex: let for $n, m \in \mathbb{N} \cup \{0\}$, $p_{n,m}(s,t) = s^n t^m$

Then $P([a,b], [c,d]) = \text{span}\{p_{n,m} : n, m \in \mathbb{N} \cup \{0\}\}$

is a point-separating subalgebra of $C([a,b] \times [c,d])$

with $1 \in P([a,b] \times [c,d])$. Hence

$$\overline{P([a,b] \times [c,d])} = C([a,b] \times [c,d]) \text{ (Unif. closure).}$$

Def'n: let (X, d) be compact metric space, $x_0 \in X$. Let

$$I(x_0) = \{f \in C(X) : f(x_0) = 0\}$$

Remarks: (i). $I(x_0)$ is closed in $(C(X), \|\cdot\|_\infty)$

let $e_{x_0} : C(X) \rightarrow \mathbb{R}$, $e_{x_0}(f) = f(x_0)$. Then

$$|e_{x_0}(f) - e_{x_0}(g)| = |f(x_0) - g(x_0)| \leq \|f - g\|_\infty, \text{ so } e_{x_0} \text{ is}$$

Lipschitz, hence cts. Also linear, $I(x_0) = e_{x_0}^{-1}(\{0\}) = \ker e_{x_0}$

(ii). $I(x_0)$ ideal in $C(X)$. i.e., $f \in C(X), g \in I(x_0) \Rightarrow fg \in I(x_0)$

Corollary (to S.W.T.)

let (X, d) be a compact metric space, $A \subseteq C(X)$

Satisfy

• A is an algebra

• A is point separating

• $A \subseteq I(x_0)$

Then $\bar{A} = I(x_0)$ in $(C(X), \|\cdot\|_\infty)$

proof: let $A_1 = \{f + \alpha 1 : f \in A, \alpha \in \mathbb{R}\}$

Then A_1 is an algebra in $C(X)$: linear subspace obvious

$$(f + \alpha 1)(g + \beta 1) = fg + \underbrace{\alpha g + \beta f}_{\in A} + \alpha\beta 1$$

$A \subseteq A_1 \Rightarrow A_1$ is point-separating

$1 \in A_1$ by design $0 \in A$.

By S.W.T., $\bar{A}_1 = C(X)$.

Now let $f \in I(x_0)$, and find $(f_n)_{n=1}^\infty \in (A_1)^\mathbb{N}$

so $\lim_{n \rightarrow \infty} f_n = f$ uniformly.

Write each $f_n = g_n + \alpha_n 1$, $g_n \in A$, $\alpha_n \in \mathbb{R}$.

Then (unif. conv $\Rightarrow$ ptwise conv):

$$\begin{aligned} 0 = f(x_0) &= \lim_{n \rightarrow \infty} f_n(x_0) = \lim_{n \rightarrow \infty} [g_n(x_0) + \alpha_n] \\ &= \lim_{n \rightarrow \infty} \alpha_n \end{aligned}$$

$$\text{Hence } \|f - g_n\|_{\infty} = \|f - f_n - \alpha_n 1\|_{\infty}$$

$$\leq \|f - f_n\|_{\infty} + \|\alpha_n 1\|_{\infty}$$

$$= \|f - f_n\|_{\infty} + |\alpha_n| \xrightarrow{n \rightarrow \infty} 0$$

$$\Rightarrow f = \lim_{n \rightarrow \infty} g_n, \text{ each } g_n \in A. \text{ So } f(x_0) \in \bar{A}.$$

$$\text{But } A \subseteq I(x_0) \text{ so } \bar{A} \subseteq \overline{I(x_0)} = I(x_0)$$

□.

$$\mathbb{C}\text{-case: } (x+iy)(a+ib) = xa - yb + i(xb + ya)$$

$$\mathbb{C} \cong \mathbb{R}^2, \text{ hence with some norm from } \mathbb{R}^2, \text{ usually } \|\cdot\|_2 = |\cdot|$$

Fact: (X, d) is a compact metric space

$C(X, \mathbb{C})$ is an algebra of functions

complete w.r.t. $\|f\|_\infty = \max_{x \in X} |f(x)|$ (module in $\mathbb{C}$).

Corollary (to S.M.T.)

let (X, d) be a compact metric space, and

$A_{\mathbb{C}} \subseteq C(X, \mathbb{C})$ satisfy

• $A_{\mathbb{C}}$ is a $\mathbb{C}$ -algebra

• $A_{\mathbb{C}}$ is point separating.

• $1 \in A_{\mathbb{C}}$

Then $\overline{A_{\mathbb{C}}} = C(X, \mathbb{C})$, in $(C(X, \mathbb{C}), \|\cdot\|_\infty)$

key idea: $\operatorname{Re} f = \frac{1}{2}(f + f^*)$, $f^*(x) = \overline{f(x)}$

$$\operatorname{Im} f = \frac{1}{2}i(f - f^*)$$

or

Def'n: let $C_0(\mathbb{R}^n) = \{f \in C(\mathbb{R}^n), \lim_{\|x\|_2 \rightarrow \infty} f(x) = 0\}$ "vanishing at ∞ "

eg. : If $p(x) = p(x_1, \dots, x_n)$ is a polynomial, $\alpha > 0$

$$f(x) = p(x) e^{-\alpha \cdot \|x\|_2^2} \text{ defines } f \in C_0(\mathbb{R}^n)$$

Q: Is $\mathcal{A} = \text{Span}\{p(\cdot) e^{-\alpha \cdot \|\cdot\|_2^2} : p \text{ poly}, \alpha \in \mathbb{R}_{>0}\}$

dense in $C_0(\mathbb{R}^n)$?

Theorem: (i). $C_0(\mathbb{R}^n) \subseteq C_b(\mathbb{R}^n)$ and is unif. closed.

(ii). (version of S.W.T.) If $\mathcal{A} \subseteq C_0(\mathbb{R}^n)$

satisfies $\cdot$ $\mathcal{A}$ is a algebra

needed at (*) $\left\{ \begin{array}{l} \cdot \mathcal{A} \text{ is point-separating} \end{array} \right.$

$\cdot \mathcal{A}$ separates points from zero: if $x \in \mathbb{R}^n$,

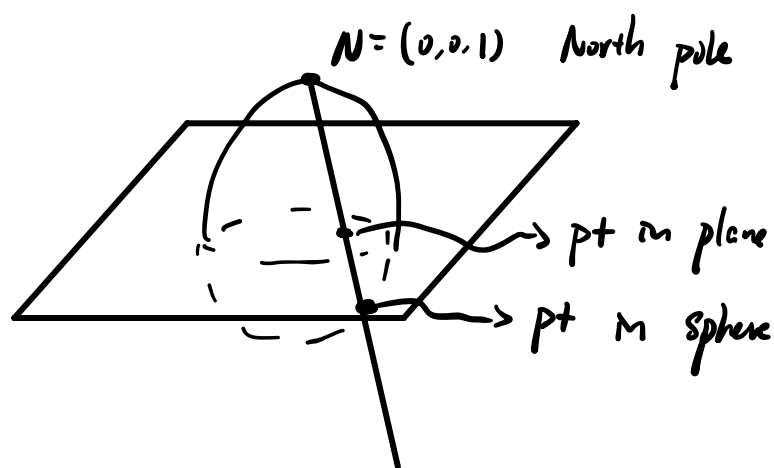
$\exists f \in \mathcal{A}$ so $f(x) \neq 0$.

Then $\overline{A} = C(\mathbb{R}^n)$ in $(C(\mathbb{R}^n), \|\cdot\|_\infty)$

proof: Use some device to get (i) and (ii).

We'll consider a version of "stereographic projection"

$$S = \{x \in \mathbb{R}^{n+1} : \|x\|_2 = 1\}$$



Define $N = (0, \dots, 0, 1)$. $\sigma: \mathbb{R}^n \rightarrow S \setminus \{N\}$

$$\sigma(x) = \left(\frac{2x_1}{\|x\|_2^2 + 1}, \dots, \frac{2x_n}{\|x\|_2^2 + 1}, \frac{\|x\|_2^2 - 1}{\|x\|_2^2 + 1} \right)$$

[compute $\|\sigma(x)\|_2 = 1$ in $\mathbb{R}^{n+1}$]

$$z: S \setminus \{N\} \rightarrow \mathbb{R}^n, \quad z(y) = \left(\frac{y_1}{1-y_{n+1}}, \dots, \frac{y_n}{1-y_{n+1}} \right)$$

check that:
 • each σ, z are cts (rational fns)

$$\bullet \quad z = \sigma^{-1}$$

$$\bullet \quad \lim_{\|x\|_2 \rightarrow \infty} \sigma(x) = N \text{ in } S.$$

$$\lim_{y \rightarrow N} \|z(y)\|_2 = \infty$$

Define $T: C(\mathbb{R}^n) \rightarrow I(N) = \{f \in C(S) : f(N) = 0\}$

$$Tf = \begin{cases} f \circ z \text{ on } S \setminus \{N\} \\ 0 \text{ at } N \end{cases}$$

$$\text{notice that } \lim_{y \rightarrow N} Tf(y) = \lim_{y \rightarrow N} f(z(y)) = \lim_{\|x\|_2 \rightarrow \infty} f(x) = 0$$

So indeed Tf is cts and $Tf \in I(N)$

Then $\bullet$ T is linear

}

- T is multiplicative i.e., $T(fg) = T(f) \cdot T(g)$
 - T is an isometry: $\|Tf\|_\infty = \|f\|_\infty$
 takes closed set $\hookrightarrow$ in $C(S)$ to closed set in $C(\mathbb{R}^n)$
 - $T(C(\mathbb{R}^n))$ is point-separating in S
- } easy, similar to $A_6 \text{ or } A_3(a)$

$[C(\mathbb{R}^n)$ separates point in $\mathbb{R}^n$ (use A from example above)]

Hence, by Corollary to S.W.T., $T(C(\mathbb{R}^n)) = I(N)$, hence closed.

(i). $T^{-1}: I(N) \rightarrow C(\mathbb{R}^n)$ is a surjection isometry,

so $C(\mathbb{R}^n) = T^{-1}(I(N))$ is closed.

(ii). Then $T(A) = \{Tf: f \in A\}$ is a point-separating (*)

algebra in $I(N)$. Again, use corollary to S.W.T.

□.

Remark (i). In $(\mathbb{R}^n, \|\cdot\|)$ (any norm), if $A \subset \mathbb{R}^n$

$\bar{A}$ is compact $\Leftrightarrow A$ is bounded (HB)

(ii). If (X, d) is an infinite compact metric space,

then for any $f \in C(X)$, any $r > 0$,

$B_c[f, r]$ is never compact. [See Ab, Q4(a) for $r \in [0, 1]$].

Lemma: Let (M, d) be a complete metric space. $A \subset M$.

Then $\bar{A}$ is compact $\Leftrightarrow A$ is totally bounded.

proof: ($\Rightarrow$) $\bar{A}$ is compact $\Rightarrow \bar{A}$ is totally bounded. if $\varepsilon > 0$

$\bar{A} \subset \bigcup_{k=1}^n B(y_k, \frac{\varepsilon}{2})$. Then, we can find

$z_k \in A \cap B(y_k, \frac{\varepsilon}{2})$ as $y_k \in \bar{A}$.

So $A \subset \bigcup_{n=1}^n B(z_n, \varepsilon)$

($\Leftarrow$): If f is totally bounded, $\varepsilon > 0$, there are $z_1, \dots, z_n \in A$

so $A \subseteq \bigcup_{k=1}^n B[z_k, \varepsilon]$. Then $\overline{A} \subseteq \bigcup_{k=1}^n B[z_k, \varepsilon]$

as a finite union of closed set is closed.

□.

Arzelà-Ascoli Theorem: let (X, d) be a compact metric space,

$\mathcal{F} \in C(X)$. Then

$\overline{\mathcal{F}}$ is compact in $(C(X), \|\cdot\|_\infty) \Leftrightarrow \mathcal{F}$ is bounded

$\mathcal{F}$ is equicontinuous

Def'n: $\mathcal{F} \subseteq C(X)$ is equicontinuous if $\forall \varepsilon > 0, \exists \delta > 0$ so

$d(x, y) < \delta$ in $X \Rightarrow |f(x) - f(y)| < \varepsilon, \forall f$ in $\mathcal{F}$

[equicontinuous = "unif. unif. cts"]

proof: ($\Rightarrow$): $\overline{\mathcal{F}}$ compact $\Rightarrow \mathcal{F}$ totally bounded

$\Rightarrow \mathcal{F}$ is totally bdd (†) $\Rightarrow \mathcal{F}$ is bdd.

By (†), given $\varepsilon > 0$, we can find $f_1, \dots, f_k$ in $\mathcal{F}$

$$\text{so } \mathcal{F} \subseteq \bigcup_{k=1}^n B_c(f_k, \frac{\varepsilon}{3}).$$

For each $k = \{1, \dots, n\}$, let $\delta_k > 0$ be so $d(x, y) < \delta_k$ in X .

we have $|f_k(x) - f_k(y)| < \frac{\varepsilon}{3}$ (compact $\Rightarrow$ cts = unif. cts)

Now let $\delta = \min\{\delta_1, \dots, \delta_n\}$, and if $f \in \mathcal{F}$, find k so

$f \in B_c(f_k, \frac{\varepsilon}{3})$, and for $d(x, y) < \delta \leq \delta_k$ in X .

$$|f(x) - f(y)| \leq |f(x) - f_k(x)| + |f_k(x) - f_k(y)| + |f_k(y) - f(y)|$$

$$< \|f - f_k\|_{\infty} + \frac{\varepsilon}{3} + \|f - f_k\|_{\infty} < \varepsilon$$

($\Leftarrow$): We shall verify total boundedness of $\mathcal{F}$.

Hence by lemma (last class), $\overline{\mathcal{F}}$ is compact in $(C(X), \|\cdot\|_\infty)$

equi-continuity. $\left\{ \begin{array}{l} \text{let } \varepsilon > 0, \text{ let us find } \delta > 0 \text{ so} \\ d(x, y) < \delta \text{ in } X \Rightarrow |f(x) - f(y)| < \frac{\varepsilon}{3}, \forall f \in \mathcal{F} \end{array} \right.$

We notice that $X = \bigcup_{x \in X} B(x, \delta)$ so for some $x_1, \dots, x_n \in X$,

$$\text{we have } X = \bigcup_{k=1}^n B(x_k, \delta)$$

Now define $T: C(X) \rightarrow \mathbb{R}^n$

$$Tf = (f(x_1), \dots, f(x_n))$$

and notice that

• T is linear

$$\bullet \|Tf\|_\infty = \max_{k=1, \dots, n} |f(x_k)| \leq \|f\|_\infty$$

so T is cts. [T has Lipschitz const. 1]

Since $\mathcal{F}$ is bdd, say $\sup_{f \in \mathcal{F}} \|f\|_{\infty} < M$ in $C(X)$.

We see that $T(\mathcal{F})$ is bounded in $\mathbb{R}^n$.

$$\|Tf\|_{\infty} \leq \|f\|_{\infty} \leq M, \quad \forall f \in \mathcal{F}. \text{ Hence}$$

$\overline{T(\mathcal{F})}$ is compact in $\mathbb{R}^n$ (HB)

Thus, $T(\mathcal{F})$ is totally bounded in $\mathbb{R}^n$

so we can find $f_1, \dots, f_m \in \mathcal{F}$ s.t.

$$T(\mathcal{F}) = \bigcup_{j=1}^m B_{\infty}(Tf_j, \frac{\varepsilon}{3}) \quad \text{["balls in } \|\cdot\|_{\infty} \text{ in } \mathbb{R}^n \text{"}]$$

Now, let $f \in \mathcal{F}$, so $Tf \in B_{\infty}(Tf_j, \frac{\varepsilon}{3})$, for some $j=1, \dots, m$

Further, if $x \in X$, $x \in B(x_k, \delta)$ for some $k=1, \dots, n$

$$|f(x) - f_j(x)| \leq \underbrace{|f(x) - f(x_k)|}_{< \varepsilon/3} + \underbrace{|f(x_k) - f_j(x_k)|}_{\leq \|Tf - Tf_j\|_{\infty}} + \underbrace{|f_j(x_k) - f_j(x)|}_{< \varepsilon/3}$$

$$\begin{aligned}
 &< \frac{2}{3}\varepsilon + \underbrace{\|Tf - Tf_k\|_\infty}_{\text{in } \mathbb{R}^n} < \frac{2}{3}\varepsilon + \frac{\varepsilon}{3} = \varepsilon. \\
 &< \frac{\varepsilon}{3} \text{ by choice of } f_j.
 \end{aligned}$$

$$\text{So } \|f - f_j\|_\infty = \max_{x \in X} |f(x) - f_j(x)| < \varepsilon, \text{ i.e., } f \in \bigcup_{j=1}^{\infty} B_c(f_j, \varepsilon)$$

□.

Peano's Theorem: Let $D \subset \mathbb{R}^2$ be open, $(t_0, y_0) \in D$.

$\bar{\Phi}: D \rightarrow \mathbb{R}$ be cts. Then

$$(IUP): \quad f'(t) = \bar{\Phi}(t, f(t)), \quad f(t_0) = y_0.$$

admits a cts sol'n on $[t_0-h, t_0+h]$ for some $h > 0$.

Remark: Uniqueness is not assured.

$$\text{Example: } f'(t) = f(t)^{\frac{1}{3}}, \quad f(0) = 0$$

$\bar{\Phi}(t, y) = y^{\frac{1}{3}}$ is cts on $\mathbb{R}^2$, but not Lipschitz in 2nd coordinate.

Sol'n: $f_0(t) = 0$, $f_1(t) = \begin{cases} (\frac{3}{2}t)^{3/2} & , t \geq 0 \\ -(\frac{3}{2}(-t))^{3/2} & , t < 0 \end{cases}$

Idea of proof of Peano:

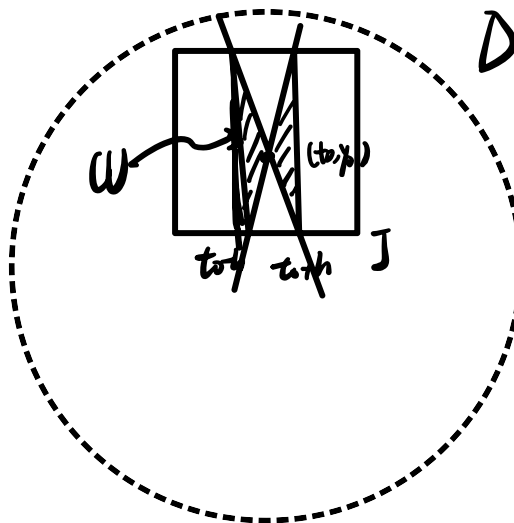
(I). Find $h > 0$. let $H, R > 0$ so

$$J = [t_0 - H, t_0 + H] \times [y_0 - R, y_0 + R] \subset D.$$

let $M = \max_{(t,y) \in J} |\tilde{F}(t,y)|$, J is compact.

let $W = \{(t,y) \in J : |y - y_0| \leq M|t - t_0|\}$,

and $h > 0$ s.t. $[t_0 - h, t_0 + h] \times \{y_0\} \subseteq W$.



(II). Find "almost soln", company a family with

unif. compact closure. Given $\varepsilon > 0$, let $\delta > 0$ s.t.

$$\|(t, y) - (t', y')\|_{\infty} < \delta \Rightarrow |\Phi(t, y) - \Phi(t', y')| < \frac{\varepsilon}{h}$$

let

$$f_{\varepsilon}(t) = \begin{cases} y_0 + \Phi(t_0, y_0) \cdot (t - t_0) & t \in [t_0, t_1] \\ f_{\varepsilon}(t_1) + \Phi(t_1, f_{\varepsilon}(t_1)) \cdot (t - t_1) & t \in [t_1, t_2] \\ \vdots \\ f_{\varepsilon}(t_{n-1}) + \Phi(t_{n-1}, f_{\varepsilon}(t_{n-1})) \cdot (t - t_{n-1}) & t \in [t_{n-1}, t_n] \end{cases}$$

where $t_0 < t_1 < \dots < t_n = t_0 + h$, $t_j - t_{j-1} < \delta$.

Check that: (1) $Tf(t) = y_0 + \int_{t_0}^t \Phi(s, f(s)) ds$

$$\|f_{\varepsilon} - Tf_{\varepsilon}\|_{\infty} < \varepsilon.$$

"approx soln"

[idea: $f_{\varepsilon}'(t) = \Phi(t, f_{\varepsilon}(t))$, $t \in (t_j, t_{j+1})$ use integral estimate].

(2). $s, t \in [t_0, t_0 + h]$, $|f_\varepsilon(s) - f_\varepsilon(t)| \leq M|s - t|$ [Indep of ε]

Now, we have built

$$\mathcal{F} = \{f_\varepsilon : \varepsilon \in (0, 1)\}$$

and (1) tell us is equi-Lipschitz.

$\Rightarrow$ equi-cts.

A.A.T. $\Rightarrow \overline{\mathcal{F}}$ is unif. compact.

(II). Extract unif. cty. subseq.

$$(f_{n_k})_{k=1}^\infty \text{ of } \mathcal{F}$$

and check $f = \lim_{k \rightarrow \infty} f_{n_k}$ satisfies IVP.

(i.e., fixed pt) $\square$.

Baire Category Theorem

Defn: Let (X, d) be a metric space

(i). A subset $N \subseteq X$ is nowhere dense

$$\text{if } (\bar{N})^\circ = \emptyset$$

(ii). A subset $M \subseteq X$ is called meager (1^{st} case)

$$\text{if } M = \bigcup_{n=1}^{\infty} N_n, \text{ where each } N_n \text{ is nowhere dense.}$$

(i.e., M is a countable union of nowhere dense sets)

(iii). A subset $A \subseteq X$ is non-meager (or 2^{nd} case)

if it is not meager.

(iv). A subset $R \subseteq X$ is residual if $X \setminus R$ is meager.

Remarks: • no-where dense $\Rightarrow$ meager

• If (X, d) is complete, then residual $\Rightarrow$ non-meager

Ex: (i) If $x_0 \in X$, $\{x_0\}$ is meyer $\Leftrightarrow x_0 \notin X'$
(i.e., isolated.)

BC7: (i). A countable intersection of dense open set is dense

(ii). $M \subseteq X$ is meyer $\Rightarrow M^\circ = \emptyset$

Pf: (ii). $M = \bigcup_{n=1}^{\infty} N_n \Rightarrow X \setminus \overline{N_n}$ is dense and open

$$\Rightarrow \bigcap_{n=1}^{\infty} (X \setminus \overline{N_n}) = X \setminus \left(\bigcup_{n=1}^{\infty} \overline{N_n} \right) \subseteq X \setminus \left(\bigcup_{n=1}^{\infty} N_n \right) \\ = X \setminus M$$

By (i), $\bigcap_{n=1}^{\infty} (X \setminus \overline{N_n})$ is dense $\Rightarrow X \setminus M$ is dense

$$\Rightarrow \forall x \in X, \varepsilon > 0, B(x, \varepsilon) \cap (X \setminus M) \neq \emptyset$$

$$\Rightarrow B(x, \varepsilon) \not\subseteq M \Rightarrow M^\circ = \emptyset.$$

Corollary: (X, d) complete, $R \in X$ residual $\Rightarrow R$ is non-meyer

Pf: R residual $\Rightarrow X \setminus R = \bigcup_{n=1}^{\infty} N_n$

Sps R is meager, then $R = \bigcup_{n=1}^{\infty} U_n$

$\Rightarrow X = \bigcup_{n=1}^{\infty} N_n \cup \bigcup_{n=1}^{\infty} U_n$, countable union of n.m. decs.

$\Rightarrow X$ is meager

By (ii), $X^o = \emptyset$

Def'n: (X, d) metric space.

closed set
↓

(i) A set $G \subseteq X$ is Gd if $G = \bigcap_{n=1}^{\infty} U_n$, U_n open

(ii) A set $F \subseteq X$ is Fd if $F = \bigcup_{n=1}^{\infty} C_n$, C_n closed.

↑
open sets